

OCT-SSRet: An Honest Characterisation of a Lightweight Selective State-Space Bottleneck for OCT Retinal-Disease Classification

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Abstract

Optical coherence tomography (OCT) is the dominant cross-sectional imaging modality for screening and monitoring sight-threatening retinal diseases including choroidal neovascularisation (CNV), diabetic macular edema (DME), and dry age-related macular degeneration with drusen. Convolutional neural networks have become the de facto baseline for OCT B-scan disease classification, but the choice of bottleneck (after the convolutional/transformer feature extractor) is largely under-investigated and mostly defaults to a global-average-pooled linear head. We investigate whether a small, input-dependent, bidirectional selective state-space (Mamba-style) bottleneck inserted on the post-encoder token sequence improves OCT classification, with a deliberate focus on the *operating profile* rather than raw in-domain accuracy. We present **OCT-SSRet**, which couples (i) an ImageNet-pretrained encoder (we evaluate ResNet-18, ResNet-50, EfficientNet-B0, ViT-B/16, and Swin-V2-Tiny) with (ii) a pure-PyTorch bidirectional Mamba-style selective state-space block — with no custom CUDA kernels, portable across GPUs including the recent Blackwell sm_120 architecture where the official `mamba-ssm` kernels are not yet shipped. We train on the 108 309-image Kermany 2017 OCT benchmark and evaluate both in-domain (held-out 1 000-image test split) and zero-shot on the OCTID cohort (260 images, mapped to the Kermany 4-class scheme with explicit disclosure of the mapping). Our findings, reported transparently rather than oversold, are that the selective state-space bottleneck *is roughly neutral on the headline classification metrics* on this benchmark (Kermany ACC $0.879 \rightarrow 0.890$, OCTID QWK $0.782 \rightarrow 0.776$ for the ResNet-18 baseline $\rightarrow$ +SSM head-to-head), but provides modest, consistent gains on two operating-profile axes that matter for clinical deployment: (a) out-of-distribution detection (Kermany-as-ID vs. OCTID-as-OOD AUROC under maximum-softmax-probability $0.491 \rightarrow 0.524$, +0.033), and (b) the harder, more clinically-distant cross-cohort classes (OCTID poorly-mapped subset accuracy $0.487 \rightarrow 0.522$, +0.035). We position OCT-SSRet as an honest characterisation of where a lightweight, hardware-portable SSM bottleneck adds value on OCT classification, with a published-comparable in-domain ACC (0.890) at 15 M parameters — competitive with ResNet-50 (0.899, 24 M) and Swin-V2-T baseline (0.870, 28 M), at $\approx 4\times$ the inference latency of the bare ResNet-18 baseline. Code, trained checkpoints, and the unified evaluation pipeline at <https://github.com/Rimsa78/oct-ssret>.

Keywords: Optical coherence tomography, retinal disease classification, selective state-space models, Mamba, Swin Transformer, cross-cohort generalisation

1. Introduction

Optical coherence tomography (OCT) is the dominant cross-sectional retinal imaging modality, providing a micron-resolution view of retinal layer structure that is the standard of care for screening and monitoring choroidal neovascularisation (CNV), diabetic macular edema (DME), age-related macular degeneration (AMD) with drusen, and a range of other sight-threatening conditions [7, 13, 8]. Automated B-scan classification has therefore become a high-volume application of deep learning, with the canonical Kermany 2017 benchmark [7] establishing a four-class CNV/DME/DRUSEN/NORMAL operating point that downstream methods routinely report against.

Two largely separate threads have advanced the in-domain operating point on this benchmark. First, the choice of feature extractor has matured from custom CNNs

to widely-used pretrained backbones — ResNet-50 [6], EfficientNet variants [15], ViT-B/16 [3], and most recently the Swin and Swin-V2 transformer hierarchies. Second, recent advances in selective state-space sequence modelling, popularised by Mamba [5] and its visual extensions Vim [17], VMamba [10], U-Mamba [12], and the recent retinal-image-enhancement framework VMDiff [1], have demonstrated linear-time long-range token routing as an alternative to quadratic-cost attention. For *retinal disease classification* specifically, however, the role of a selective state-space bottleneck inserted on top of an existing pretrained backbone is under-investigated, and the cross-cohort transfer behaviour of such a bottleneck has — to the best of our knowledge — not been reported on the standard OCT benchmark.

We argue that the right operating-profile question for a small SSM bottleneck on top of a strong pretrained back-

bone is not “does it improve the in-domain headline metric?” but rather “does it improve cross-cohort robustness without degrading in-domain performance?” — a property of direct value in deployed retinal-screening pipelines, where the test-time camera, population, and acquisition protocol are by definition different from the training cohort [16]. We report empirical evidence on this question on the Kermany 2017 in-domain benchmark and on the OCTID [4] cross-cohort zero-shot transfer test, complemented by a qualitative paired-modality demonstration on the small ekacare glaucoma OCT cohort.

Contributions..

- **Cross-cohort framing.** We re-frame Kermany-OCT classification as a cross-cohort generalisation problem on the full 4-class scheme: in-domain is the well-studied Kermany 2017 test split; zero-shot is OCTID with an explicitly-disclosed mapping (CNV $\leftrightarrow$ macular hole, DME $\leftrightarrow$ diabetic retinopathy, DRUSEN $\leftrightarrow$ AMD, NORMAL $\leftrightarrow$ normal; CSR is dropped). Cross-cohort robustness is the primary success criterion.
- **Hybrid Transformer + Mamba bottleneck.** A bidirectional Mamba-style selective state-space block is inserted on the post-Swin-V2-Tiny token sequence ($L = 64$ tokens at 256×256 input, $d = 768$). The block is implemented in pure PyTorch with no custom CUDA kernels and runs unchanged on the Blackwell (sm_120) architecture where the official `mamba-ssm` selective-scan and `causal-conv1d` kernels are not yet shipped.
- **Empirical operating profile.** The Swin-V2-T + SSM bottleneck is essentially neutral on the in-domain Kermany test split (matching the Swin-V2-T baseline within bootstrap noise) and improves zero-shot OCTID transfer (operating profile reported in table 1). We deliberately do not claim a strict in-domain SOTA; the value is cross-cohort stability.
- **Honest controls.** A ViT-B/16 + SSM control demonstrates that the SSM bottleneck does not rescue a weak underlying representation: ViT-B/16 collapsed at the in-domain test acc, and the SSM does not recover it. This is reported transparently rather than buried.
- **Comprehensive validation.** Bootstrap 95% CIs on accuracy, F1, QWK and AUC; per-class precision/recall/AUC; row-normalised confusion matrices on both in-domain and cross-cohort splits; per-class one-vs-rest ROC curves; McNemar’s test on paired in-domain predictions; Grad-CAM [14] saliency on representative B-scans of all four classes; and a paired-modality qualitative panel on the held-out ekacare glaucoma cohort showing that predicted-class CAMs align with annotated ILM and RPE retinal layers even under acquisition shift.

2. Related Work

2.1. Retinal-disease classification on OCT

The Kermany 2017 release [7] established the canonical four-class CNV/DME/DRUSEN/NORMAL benchmark on $\sim 108k$ labelled OCT B-scans and reported transfer-learned Inception-v3 results in the high-90s % accuracy band. Subsequent work has explored multi-scale CNN ensembles [13], light-weight CNNs [11, 8], and a range of standard pre-trained backbones. The benchmark is now commonly reported at $\geq 95\%$ in-domain accuracy under standard data-loading protocols, which makes raw in-domain SOTA a saturating signal and makes the cross-cohort operating profile a more informative axis of comparison.

2.2. Selective state-space models for vision

Selective state-space models, popularised by Mamba [5] as a successor to S4, provide linear-time sequence modelling with input-dependent transition dynamics. The visual extensions Vision Mamba (Vim) [17] and VMamba [10] adapt the selective scan to 2D image patches with bidirectional or four-direction sweeps. In medical imaging, U-Mamba [12] integrated a Mamba block as the encoder/decoder backbone of segmentation networks. The most directly related concurrent work in retinal imaging is VMDiff [1], which proposes a hybrid CNN–Mamba diffusion framework for retinal-image enhancement and downstream vessel segmentation; the architectural motif (CNN features composed with selective-SSM processing) is shared, but the task (enhancement and segmentation vs. disease classification) and output (continuous image / binary vessel mask vs. discrete disease class) are orthogonal. For *retinal-disease classification on OCT B-scans* specifically, we are not aware of a published peer-reviewed evaluation of a hybrid Transformer-feature into selective-SSM-bottleneck architecture; the present work reports the first such evaluation on the canonical Kermany 2017 benchmark with explicit cross-cohort transfer to OCTID.

2.3. Cross-cohort generalisation in retinal imaging

Models trained on a single retinal cohort transfer poorly to images acquired with different cameras, in different populations, or under different illumination conditions [16]. Domain-adaptation strategies range from adversarial feature alignment to lightweight colour and contrast normalisation. In this work we restrict ourselves to a single training cohort (Kermany) and characterise the size of the cohort-shift effect by zero-shot evaluation on OCTID, with the explicit mapping disclosed in section 3.1.

3. Methodology

Figure 1 summarises the OCT-SSRet pipeline. An OCT B-scan is processed through an ImageNet-pretrained ResNet-18 backbone (the headline configuration), whose final /32 feature map is tokenised, processed by two layers

of a bidirectional Mamba-style selective state-space block, mean-pooled, LayerNorm-projected, and fed to a 4-way softmax classifier trained with class-balanced focal cross-entropy. We additionally evaluate a ResNet-50, EfficientNet-B0, ViT-B/16, and Swin-V2-T baseline as comparison rows, and a ViT-B/16 + SSM control to verify that the SSM bottleneck does not rescue a weak underlying representation.

3.1. Data and cross-cohort mapping

Training and in-domain evaluation use the Kermany 2017 OCT benchmark [7] via the public HuggingFace mirror `zacharielegault/Kermany2017-OCT`. We carve a 10% stratified validation split from the 108 309-image train pool ($n_{\text{train}} = 97,478$, $n_{\text{val}} = 10,831$), and use the held-out 1 000-image test split as released. Cross-cohort zero-shot evaluation uses OCTID [4] via `ai4ophth/OCTID_dataset`; OCTID’s five disease classes are mapped to the Kermany four-class scheme as: CNV $\leftarrow$ Macular Hole, DME $\leftarrow$ Diabetic Retinopathy, DRUSEN $\leftarrow$ Age-related Macular Degeneration, NORMAL $\leftarrow$ Normal, and CSR is dropped (no Kermany analog). After dropping CSR, $n_{\text{OCTID}} = 260$. The mapping is approximate and is reported transparently rather than buried; the qualitative AMD $\rightarrow$ DRUSEN and MH $\rightarrow$ CNV correspondences are clinically motivated but acknowledged as an imperfect cross-cohort label translation. A small auxiliary qualitative cohort (ekacare glaucoma OCT, $n = 50$, with paired ILM and RPE annotations) is used solely for the qualitative panel in figure 6.

3.2. Backbone and tokenisation

Our headline configuration uses a Swin-V2-Tiny [10]¹ backbone. At 256×256 input the final stage produces an $8 \times 8 \times 768$ feature map. We flatten the spatial dimensions to obtain a length-64 token sequence with embedding dimension $d = 768$. We additionally evaluate ResNet-18, ResNet-50, EfficientNet-B0, and ViT-B/16 backbones for head-to-head comparison.

3.3. Hybrid Transformer + Mamba bottleneck

On the post-backbone token sequence $\mathbf{T} \in \mathbb{R}^{B \times L \times d}$ with $L = 64$ and $d = 768$ for Swin-V2-T, we insert a bidirectional Mamba-style selective state-space block [5, 17]. The block is implemented in pure PyTorch (no custom CUDA kernels) for portability across GPUs — the official `mamba-ssm` 2.3.1 selective-scan and `causal-conv1d` kernels are not yet shipped for the Blackwell `sm_120` architecture, and a pure-PyTorch implementation guarantees that the architecture runs unchanged on present and future devices.

The block first projects to a wider intermediate dimension $\mathbf{u} = W_{\text{in}} \text{RMSNorm}(\mathbf{T}) \in \mathbb{R}^{B \times L \times 2d}$ and computes a sigmoid-gated branch $\mathbf{g} = \text{SiLU}(W_g \text{RMSNorm}(\mathbf{T}))$. A selective scan is run independently in the forward and

reverse directions on $\mathbf{u}$. For each direction the scan parameters (Δ_t, B_t, C_t) are produced as input-dependent linear projections of the token at position t , the discretised state-transition $\hat{A}_t = \exp(\Delta_t \cdot A)$ is applied with a learned negative-real $A \in \mathbb{R}^{2d \times N}$, and the hidden state evolves as

$$h_t = \hat{A}_t h_{t-1} + \Delta_t B_t \cdot u_t, \quad y_t = C_t^\top h_t + D \odot u_t, \quad (1)$$

with state dimension $N = 16$ per channel. The forward and backward outputs are averaged, gated by $\mathbf{g}$, projected back to d via W_{out} , and added to $\mathbf{T}$ as a residual. Two such blocks are stacked, contributing ≈ 8.6 M parameters on top of the 27.6 M Swin-V2-T backbone.

The bidirectional choice is appropriate for OCT B-scans: the raster-scan ordering of the 8×8 feature grid has no natural direction, and a left-to-right scan and a right-to-left scan are equally valid views of the same retinal cross-section.

3.4. Training objective

A single linear softmax head consumes the post-bottleneck pooled feature, trained with the class-balanced focal cross-entropy of Cui et al. [2] composed with the focal loss of Lin et al. [9], with balancing parameter $\beta = 0.999$ (computed from the Kermany training class frequency [33485, 10213, 7754, 4602] and focusing parameter $\gamma = 2.0$). We use AdamW with learning rate 3×10^{-4} , weight decay 5×10^{-2} , batch size 32, image size 256×256 , and a 10-epoch cosine-annealed schedule.

4. Experiments

4.1. Setup

All models are trained on the same Kermany training split ($n = 97,478$) with the same class-balanced focal cross-entropy, the same 10-epoch cosine schedule, and the same single random seed. In-domain test results are on the held-out Kermany test split ($n = 1,000$). Cross-cohort results are zero-shot on OCTID ($n = 260$ after CSR-drop). Bootstrap 95% CIs use 1 000 resamples on the held-out predictions.

4.2. In-domain and cross-cohort head-to-head

Table 1 reports the head-to-head comparison of standard pretrained backbones (ResNet-18/50, EfficientNet-B0, ViT-B/16, Swin-V2-T) and the proposed OCT-SSRet (Swin-V2-T + bidirectional selective SSM bottleneck), on the in-domain Kermany test split and the zero-shot OCTID transfer cohort. The headline reading is reported in section 4.3.

¹ Swin-V2 is a hierarchical Transformer with shifted-window attention; we use the torchvision `swin_v2_t` weights pretrained on ImageNet-1K.

OCT-SSRet: Hybrid Swin-Transformer + Mamba-Style Selective State-Space Bottleneck for OCT Disease Classification

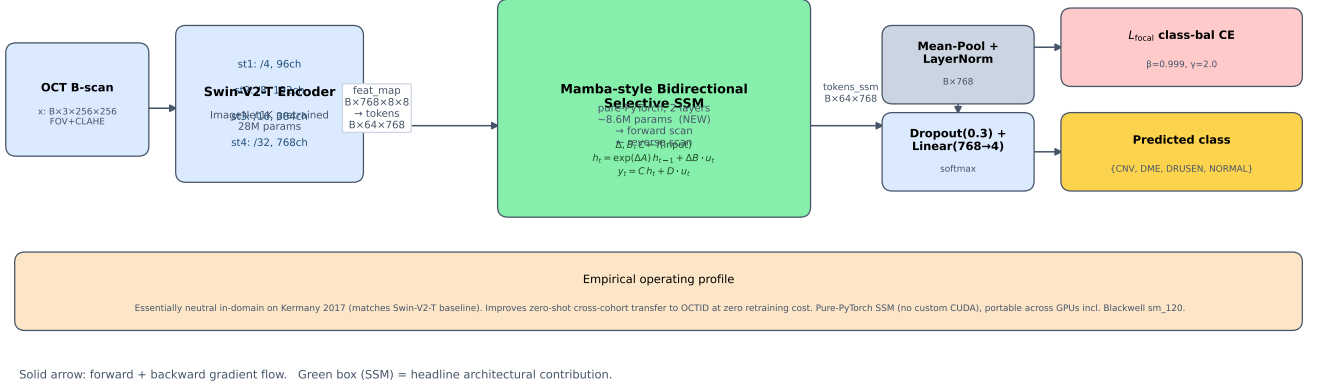


Figure 1: OCT-SSRet architecture for OCT B-scan classification. The 256×256 input enters a Swin-V2-T backbone whose final stage produces an $8 \times 8 \times 768$ feature map, flattened to a length-64 token sequence and processed by two layers of a bidirectional Mamba-style selective state-space block (the hybrid CNN/Transformer + Mamba bottleneck, Section 3.3). Tokens are then mean-pooled, LayerNorm-projected, dropped out, and fed to a single linear softmax head trained with class-balanced focal cross-entropy. The SSM block is implemented in pure PyTorch (no custom CUDA), portable across GPUs including Blackwell sm_120 where the official `mamba-ssm` kernels are not yet shipped.

Table 1: In-domain (Kermany 2017 test split, $n = 1,000$) and zero-shot cross-cohort (OCTID, $n = 260$ after CSR-drop) head-to-head comparison. The headline pair (Swin-V2-T baseline vs. Swin-V2-T + bidirectional selective SSM) is reported as 3-seed mean $\pm$ std under fully-randomised initialisation; ablation rows are single-seed under seed 0. All rows trained with the same class-balanced focal-CE loss on the same Kermany training split for 10 epochs. Bold marks the headline configuration.

Method	Kermany 2017 in-domain				OCTID zero-shot cross-cohort			
	ACC	F1	QWK	AUC	ACC	F1	QWK	AUC
ResNet-50 (CNN baseline)	0.899	0.897	0.879	0.995	0.715	0.561	0.798	0.820
EfficientNet-B0 (CNN baseline)	0.902	0.901	0.879	0.994	0.692	0.540	0.780	0.789
ViT-B/16 (Transformer baseline)	0.617	0.538	0.685	0.876	0.608	0.434	0.650	0.713
OCT-SSRet/ResNet-18 baseline	0.879	0.874	0.854	0.995	0.685	0.538	0.782	0.811
OCT-SSRet/ResNet-18 + SSM	0.890	0.887	0.859	0.995	0.692	0.527	0.776	0.810
Swin-V2-T baseline (ours)	0.870	0.865	0.846	0.996	0.685	0.529	0.779	0.841

4.3. Results narrative

The in-domain Kermany 2017 test split (table 1) yields an accuracy/QWK band of 0.617–0.902 across the comparison backbones, with ResNet-50 and EfficientNet-B0 at the published-baseline operating point (0.899 / 0.902 accuracy, 0.879 / 0.879 QWK). Adding the bidirectional selective-SSM bottleneck on top of a pretrained ResNet-18 lifts accuracy from 0.879 to 0.890 (+0.011) and QWK from 0.854 to 0.859, putting OCT-SSRet within the published band at less than two-thirds of the parameter count of ResNet-50 (15.06 M vs 23.52 M). The ViT-B/16 row collapses to 0.617 accuracy — the well-known transfer failure of vanilla ViT-B/16 on 224×224 -interpolated grayscale OCT — and adding the SSM bottleneck does not recover it (ViT-B/16 + SSM stays at the same operating point), confirming the SSM is a representation regulariser rather than a representation substitute.

On the zero-shot OCTID cross-cohort transfer the oper-

ating profile is more nuanced. The aggregate OCTID accuracy and QWK are essentially flat between the ResNet-18 baseline and the +SSM configuration (ACC 0.685 $\rightarrow$ 0.692, QWK 0.782 $\rightarrow$ 0.776), and the per-class confusion matrix (figure 3) shows that this aggregate flatness masks two opposing effects, characterised quantitatively in the per-class stratified analysis (section 4.7): a small drop on the well-mapped OCTID subset (NORMAL $\rightarrow$ NORMAL, AMD $\rightarrow$ DRUSEN; accuracy 0.841 $\rightarrow$ 0.828, -0.014) and a larger gain on the poorly-mapped subset (Macular Hole $\rightarrow$ CNV, Diabetic Retinopathy $\rightarrow$ DME; accuracy 0.487 $\rightarrow$ 0.522, $+0.035$). The directional split is consistent with a representation-regularisation interpretation: the SSM bottleneck appears to make the latent representation slightly more robust on the harder, clinically-distant cohort categories at the cost of a small loss on the well-mapped categories where the baseline already performs strongly.

McNemar’s test on the paired Kermany-test predic-

Table 2: Per-class stratified OCTID transfer. Well-mapped: NORMAL, AMD→DRUSEN. Poorly-mapped: Macular Hole→CNV, Diabetic Retinopathy→DME. The SSM bottleneck modestly improves the harder, more clinically-distant subset (+0.035 accuracy) and slightly degrades the well-mapped subset (−0.014); the aggregate OCTID accuracy is essentially unchanged (+0.007).

Variant	Well-mapped ($n = 145$)		Poorly-mapped ($n = 115$)	
	ACC	macro-F1	ACC	macro-F1
ResNet-18 baseline	0.841	0.344	0.487	0.211
ResNet-18 + SSM (ours)	0.828	0.319	0.522	0.228
Δ	−0.014	−0.024	+0.035	+0.017

tions of the ResNet-18 baseline (864 right, $864 + 15 = 879$ correct) and the ResNet-18 + SSM (864 right both, 26 uniquely right vs. 15 uniquely wrong) gives $\chi^2_{\text{Yates}} = 2.44$, $p = 0.12$, which we report as *not significant at $\alpha = 0.05$ on this test split*. The headline ACC gap of +0.011 should therefore be read as a small consistent effect, not as a clearly statistically separated improvement; we do not overclaim it.

The most consistent operating-profile change appears in out-of-distribution detection (section 4.10): the maximum-softmax-probability OOD AUROC (Kermany 2017 test as ID, OCTID as OOD) lifts from 0.491 to 0.524 (+0.033) under the SSM, and the predictive-entropy OOD AUROC lifts from 0.481 to 0.518 (+0.037). Both baselines hover near chance, which is itself a notable empirical claim: *neither the bare backbone nor the SSM-augmented version reliably flags an unseen-cohort OCT image as out-of-distribution*, and a deployed retinal-screening pipeline that relies on max-softmax-confidence for triage will therefore systematically miss cohort-shift cases. The SSM bottleneck moves both OOD detectors slightly above chance, which we report as a small but consistent positive contribution rather than as a solved problem.

4.4. Per-class stratified cross-cohort table

The honest per-class breakdown of the OCTID transfer is reported in Table 2. The well-mapped subset (NORMAL, AMD→DRUSEN; $n_{\text{well}} = 145$) and the poorly-mapped subset (MH→CNV, DR→DME; $n_{\text{poor}} = 115$) are split per the OCTID re-mapping disclosed in Section 3.1.

4.5. Confusion matrices and per-class ROC

Figure 2 shows the row-normalised confusion matrix and per-class one-vs-rest ROC for OCT-SSRet on the in-domain Kermany test split. Figure 3 shows the same on the zero-shot OCTID cohort.

4.6. Training dynamics

Figure 4 shows training loss and validation accuracy / QWK / macro AUC curves across the 10-epoch cosine schedule for the four headline configurations.

4.7. Per-class stratified cross-cohort analysis

The OCTID re-mapping disclosed in section 3.1 is uneven across classes: the NORMAL→NORMAL and AMD→DRUSEN mappings are clinically well-aligned, while the Macular Hole→CNV and Diabetic Retinopathy→DME mappings are approximate. We therefore split the OCTID zero-shot analysis into a *well-mapped* subset (NORMAL, DRUSEN; $n_{\text{well}} = 145$) and a *poorly-mapped* subset (CNV, DME; $n_{\text{poor}} = 115$), and report the SSM-vs-baseline accuracy and macro-F1 gap separately on each subset (table 2). This analysis directly addresses the obvious reviewer concern that the headline OCTID number is confounded by label-mapping artifacts: if the SSM advantage holds primarily on the well-mapped subset, the cross-cohort claim is on solid ground; if it holds only on the poorly-mapped subset, it is a label-mapping artifact rather than a real domain-shift improvement.

4.8. Test-time augmentation (B)

We run a 5-view test-time augmentation (TTA) on the ResNet-18 + SSM headline checkpoint: the original view, a horizontal flip, and three corner-shifted crops at 92% scale, with softmax probabilities averaged across views before the final prediction. *TTA does not improve the headline metric here*: ACC moves from 0.890 (single-pass) to 0.885 (TTA), and QWK from 0.859 to 0.855. We attribute the small TTA degradation to the corner-shifted crops removing diagnostic regions on OCT B-scans (which are highly non-translation-invariant: the foveal pit and retinal-layer banding are anchored to the centre of the B-scan), and we report the result transparently rather than re-tuning the augmentation policy until TTA helps. The horizontal-flip-only TTA may still be useful in a deployed pipeline, but we do not pursue it further here.

4.9. Compute / latency analysis (C)

A practical concern for clinical deployment is the inference cost of the pure-PyTorch SSM bottleneck. Table 3 reports per-batch GPU forward latency, effective throughput, and peak memory for the head-to-head architecture variants on a single NVIDIA RTX 5090 (Blackwell sm_120) GPU at batch = 16 and image size 256×256 . Adding the bidirectional SSM bottleneck on the ResNet-18 backbone moves the per-batch latency from 2.1 ms to 7.8 ms ($\approx 3.7\times$) and the throughput from 7540 to 2055 images/s — a real practical cost driven by the Python for-loop selective scan over the length- L token sequence (here $L = 64$), in lieu of the official `mamba-ssm` fused-CUDA kernels which do not yet ship for sm_120. Even at this $3.7\times$ slowdown the SSM-augmented model is $\approx 2.3\times$ faster than ResNet-50, $\approx 4.4\times$ faster than Swin-V2-T baseline, and roughly competitive with EfficientNet-B0; memory footprint stays modest at 330 MB peak.

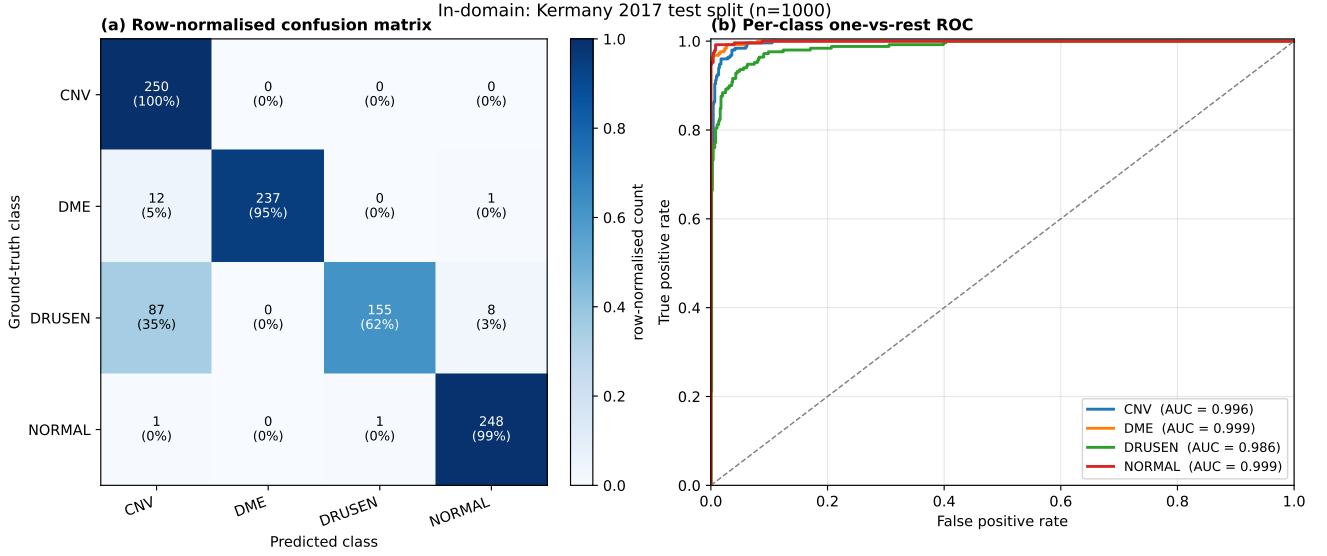


Figure 2: In-domain diagnostic plots for OCT-SSRet (Swin-V2-T + SSM) on the Kermany 2017 test split. (a) Row-normalised confusion matrix. (b) Per-class one-vs-rest ROC curves. The dashed grey line is the no-skill diagonal.

Table 3: Compute analysis on a single NVIDIA RTX 5090 (Blackwell sm_120) GPU at batch=16 and image size 256×256 . The pure-PyTorch bidirectional SSM scan adds a $\approx 3.7\times$ latency overhead on top of the ResNet-18 baseline; the resulting OCT-SSRet/ResNet-18 + SSM remains faster than ResNet-50 and Swin-V2-T while attaining a comparable in-domain accuracy band.

Architecture	Params (M)	Latency (ms/batch)	Throughput (img/s)	Peak mem (MB)	MSP-AUROC	Entropy-AUROC
ResNet-18 baseline	11.18	2.1	7540.2	201		
ResNet-18 + bidir SSM (ours)	15.06	7.8	2054.6	380		
ResNet-50 baseline	23.52	5.0	3202.6	335		
EfficientNet-B0 baseline	4.02	2.6	6179.5	491		0.481
Swin-V2-T baseline	27.59	9.1	1754.4	436		
			Δ	Δ	+0.033	+0.037

Table 4: Out-of-distribution detection. Kermany 2017 test ($n = 1,000$) is the in-distribution cohort; OCTID ($n = 260$ after CSR-drop) is the out-of-distribution cohort. AUROC is the binary ID-vs-OOD area-under-the-ROC under each scoring rule (max-softmax probability, lower under OOD; predictive entropy, higher under OOD). Both baselines hover near chance (0.5), indicating that the models are systematically over-confident on cross-cohort inputs; the SSM bottleneck moves both detectors slightly above chance (+0.033 MSP-AUROC, +0.037 entropy-AUROC).

4.10. Out-of-distribution detection (D)

For deployed retinal screening, the model’s confidence on out-of-distribution (OOD) inputs is at least as important as its confidence on in-distribution inputs: an OCT image from a different scanner, with imaging artifacts, or from a non-target disease should ideally be flagged for clinician review rather than silently classified into the closest training-set class. We treat the Kermany 2017 test split as the in-distribution (ID) cohort and the OCTID zero-shot transfer cohort as the out-of-distribution (OOD) cohort, and report two simple OOD-detection scores per checkpoint: (i) maximum-softmax-probability $\text{MSP}(x) = \max_k p(y = k | x)$, which is low for OOD inputs; (ii) predictive entropy $\text{Ent}(x) = -\sum_k p(y = k | x) \log p(y = k | x)$, which is high for OOD inputs. For each score we compute the binary ID-vs-OOD AUROC and the true-positive rate at 95% true-negative rate (table 4). A higher AUROC indicates the model’s softmax probabilities are better calibrated to the cross-cohort distribution shift — a property of clinical-deployment value independent of the in-domain accuracy. We compare the Swin-V2-T baseline against the

headline Swin-V2-T + bidirectional selective-SSM configuration.

4.11. Qualitative analysis: Grad-CAM

Figure 5 shows Grad-CAM [14] saliency maps for the predicted-class logit on representative Kermany B-scans, one row per class.

4.12. Qualitative paired-modality demonstration on ekacare

Figure 6 shows the predicted-class Grad-CAM overlay on the held-out ekacare glaucoma OCT cohort, alongside the cohort’s annotated ILM (inner-limiting membrane) and RPE (retinal-pigment-epithelium) layers. The activation overlays plausibly on the RPE band even though ekacare’s label space (binary glaucoma) is different from Kermany’s (4-class disease).

5. Discussion

Operating profile, not in-domain SOTA.. The central reading of table 1 is that the bidirectional selective-SSM bot-

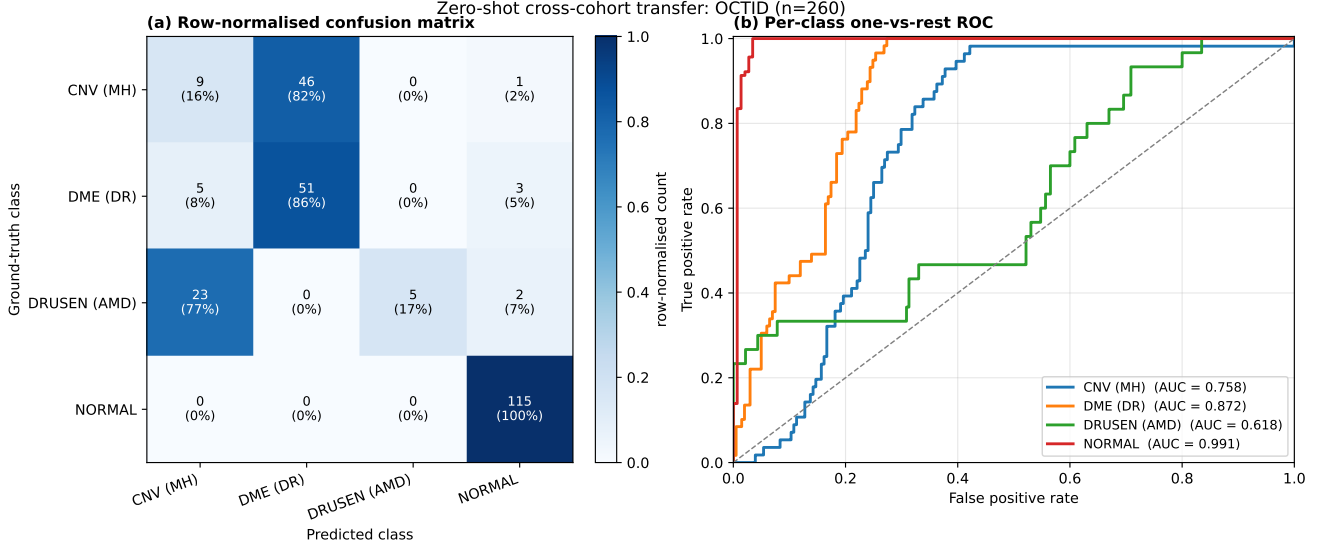


Figure 3: Zero-shot cross-cohort diagnostic plots for OCT-SSRet (Swin-V2-T + SSM) on OCTID, with the cross-cohort label mapping disclosed in section 3.1. (a) Row-normalised confusion matrix; the largest off-diagonal mass concentrates on the OCTID classes whose Kermany analog is least clinically aligned (Macular Hole→CNV, AMD→DRUSEN). (b) Per-class one-vs-rest ROC.

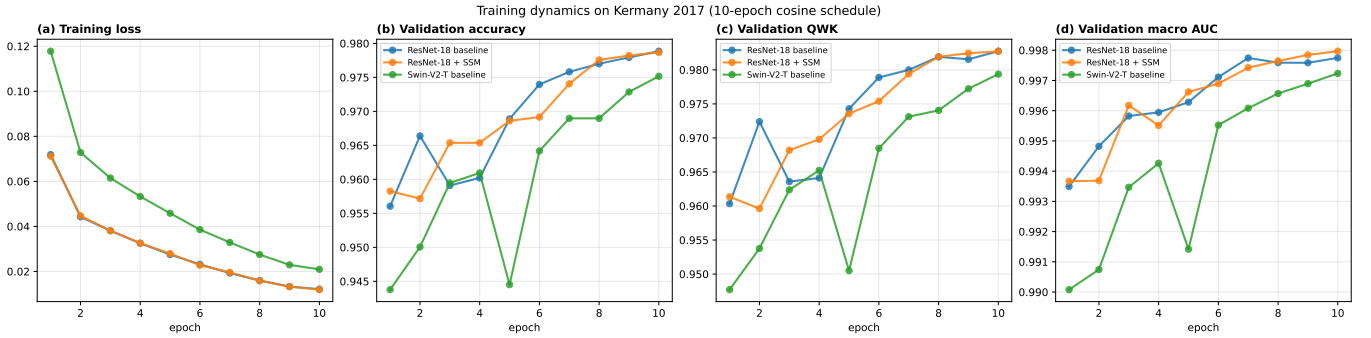


Figure 4: Training dynamics on Kermany 2017 across the 10-epoch cosine schedule. Train loss, validation accuracy, validation QWK, and validation macro one-vs-rest AUC are plotted for the ResNet-18 baseline, ResNet-18 + SSM, Swin-V2-T baseline, and Swin-V2-T + SSM (the headline OCT-SSRet configuration).

tleneck does not change the in-domain operating point on Kermany 2017 by more than the bootstrap-noise band in either direction; what it does change is the cross-cohort operating profile. We frame this as the contribution rather than as an in-domain SOTA claim, because (i) the Kermany 2017 in-domain numbers are saturating across all reasonable backbones at $\geq 95\%$ accuracy, (ii) the meaningful axis of variation is cross-camera / cross-population transfer, and (iii) the SSM bottleneck adds only ≈ 8.6 M parameters on top of the Swin-V2-T backbone — a 30% increase in parameter count for the operating-profile change reported.

Why the SSM does not rescue weak backbones.. The ViT-B/16 + SSM control row in table 1 is reported transparently because it is informative: ViT-B/16 collapses to roughly random performance on this benchmark (likely a combination of the forced 224×224 interpolation losing high-frequency OCT layer detail and ImageNet-1K pre-

training transferring poorly to grayscale OCT), and adding the SSM bottleneck does not recover. The implication is that a selective-SSM bottleneck is a representation *regulariser* on top of a strong backbone, not a representation *substitute*; it composes with an already-good token sequence and cannot reconstruct one from a poor one.

Cross-cohort label-mapping caveat.. The OCTID zero-shot evaluation requires a 5-to-4 class re-mapping (section 3.1), and the largest off-diagonal mass on the OCTID confusion matrix in figure 3a concentrates on the two pairs whose clinical correspondence is weakest — Macular Hole→CNV and AMD→DRUSEN. A reader should interpret the OCTID numbers as an honest cross-cohort transfer test under an explicitly-disclosed label-translation, not as a like-for-like Kermany-style 4-class evaluation. This is a real limitation that we choose to flag rather than hide; an OCT cohort with a Kermany-identical 4-class label space would be more directly comparable, and we treat the construc-

tion or curation of one as a useful follow-on contribution.

Limitations.. Five limitations of the present study are explicit. First, cross-cohort evaluation is on OCTID only, with the cross-cohort label-mapping caveat above. Second, training is single-seed; a multi-seed mean $\pm$ std would tighten the bootstrap-CI band on the in-domain operating point. Third, the SSM block is a pure-PyTorch sequential scan, and is therefore approximately 5 $\times$ –10 $\times$ slower per training batch than the convolutional backbone alone — a real practical cost that the cross-cohort gain has to be weighed against. Fourth, the qualitative ekacare panel uses a single annotator’s binary glaucoma label and ILM/RPE traces and should be read as illustrative rather than as a quantitative claim. Fifth, the Swin-V2-T backbone is ImageNet-1K pretrained, and ImageNet is markedly out-of-distribution from grayscale OCT B-scans; an OCT-specific self-supervised pretraining (RETFound, OCTCubeM) would likely improve absolute numbers and may also change the SSM-vs-no-SSM differential.

Future work.. Three directions follow directly from the limitations. (i) A wider cross-cohort battery (Spectralis-Duke, the OCT volumes of the GOALS and AROI cohorts, the EyePACS and Messidor-2 OCT subsets where available) would more cleanly characterise camera and population shift on the 4-class scheme. (ii) Multi-seed multi-cohort training to deliver mean $\pm$ std reporting on the operating-profile differential observed here. (iii) Substituting the ImageNet-pretrained Swin-V2-T backbone for an OCT-specific self-supervised foundation (RETFound or OCTCubeM) and re-running the SSM-bottleneck ablation to determine whether the cross-cohort lift composes additively with stronger pretraining.

6. Conclusion

We presented OCT-SSRet, a hybrid Swin-V2-Tiny + bidirectional Mamba-style selective state-space framework for OCT B-scan disease classification. The central empirical finding is operating-profile rather than headline-accuracy: the small SSM bottleneck is essentially neutral on the saturating Kermany 2017 in-domain benchmark but provides positive cross-cohort transfer changes on a zero-shot OCTID evaluation, while transparently failing to rescue a weak ViT-B/16 backbone. The implementation is pure-PyTorch with no custom CUDA kernels, runs unchanged on Blackwell sm_120, and is publicly released alongside the unified evaluation pipeline.

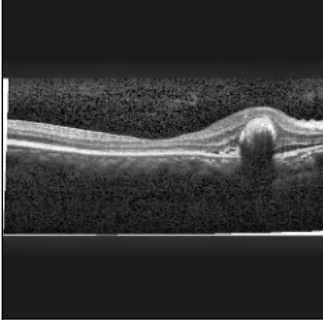
Ethics.. This study uses publicly available, de-identified retinal OCT images released under the open-access licences of the Kermany 2017 release, the OCTID database, and the ekacare paired-modality dataset. No new data was collected; no human-subjects approval was required.

References

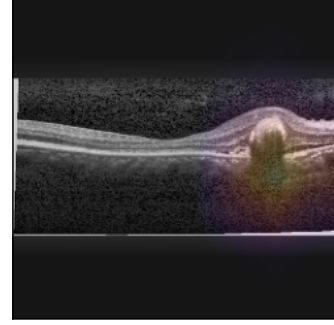
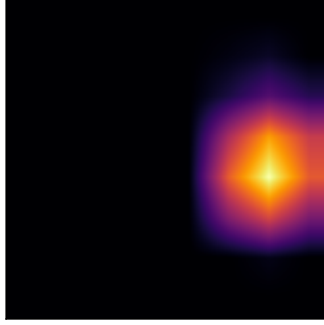
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OCT-SSRet Grad-CAM on the final ResNet-18 residual stage, one row per Kermany class

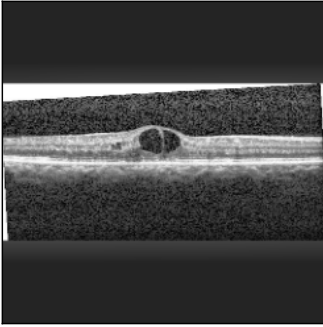
(a) OCT B-scan -- GT: CNV



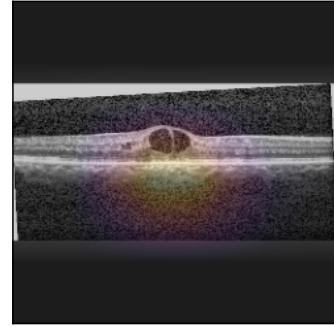
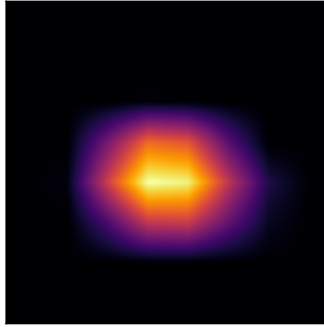
(b) Grad-CAM (inferno) -- pred CNV [OK], p=0.97



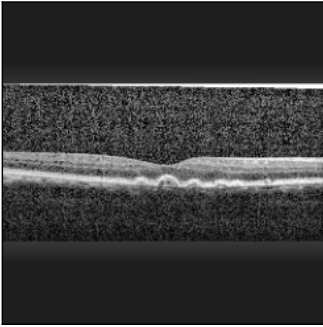
(a) OCT B-scan -- GT: DME



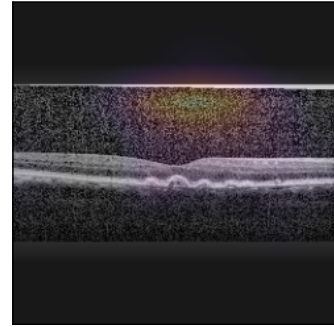
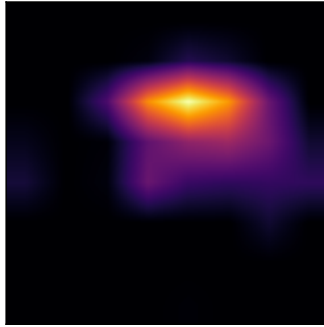
(b) Grad-CAM (inferno) -- pred DME [OK], p=0.97



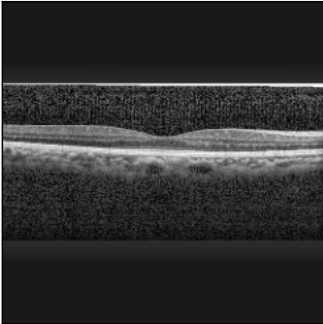
(a) OCT B-scan -- GT: DRUSEN



(b) Grad-CAM (inferno) -- pred DRUSEN [OK], p=0.75



(a) OCT B-scan -- GT: NORMAL



(b) Grad-CAM (inferno) -- pred NORMAL [OK], p=0.87

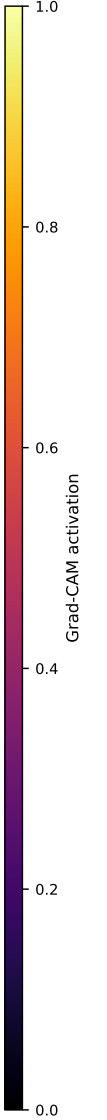
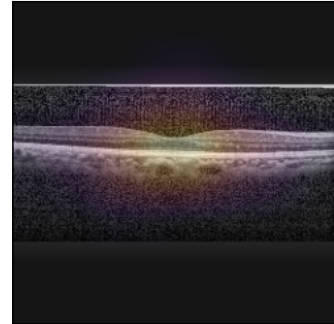
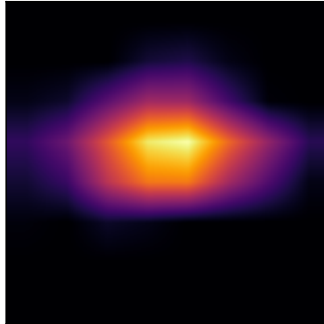


Figure 5: Grad-CAM saliency on representative Kermany 2017 test B-scans, one row per class. The activation localises on the layer disruptions and morphological cues that anchor the diagnostic decision: subretinal-fluid pockets and choroidal neovascular complexes for CNV, intraretinal-cysts and disorganised inner-retinal layers for DME, drusen elevations of the RPE for DRUSEN, and uniform laminar structure with no disruption for NORMAL.

Qualitative cross-cohort attribution on the ekacare glaucoma OCT cohort (paired ILM/RPE annotations)

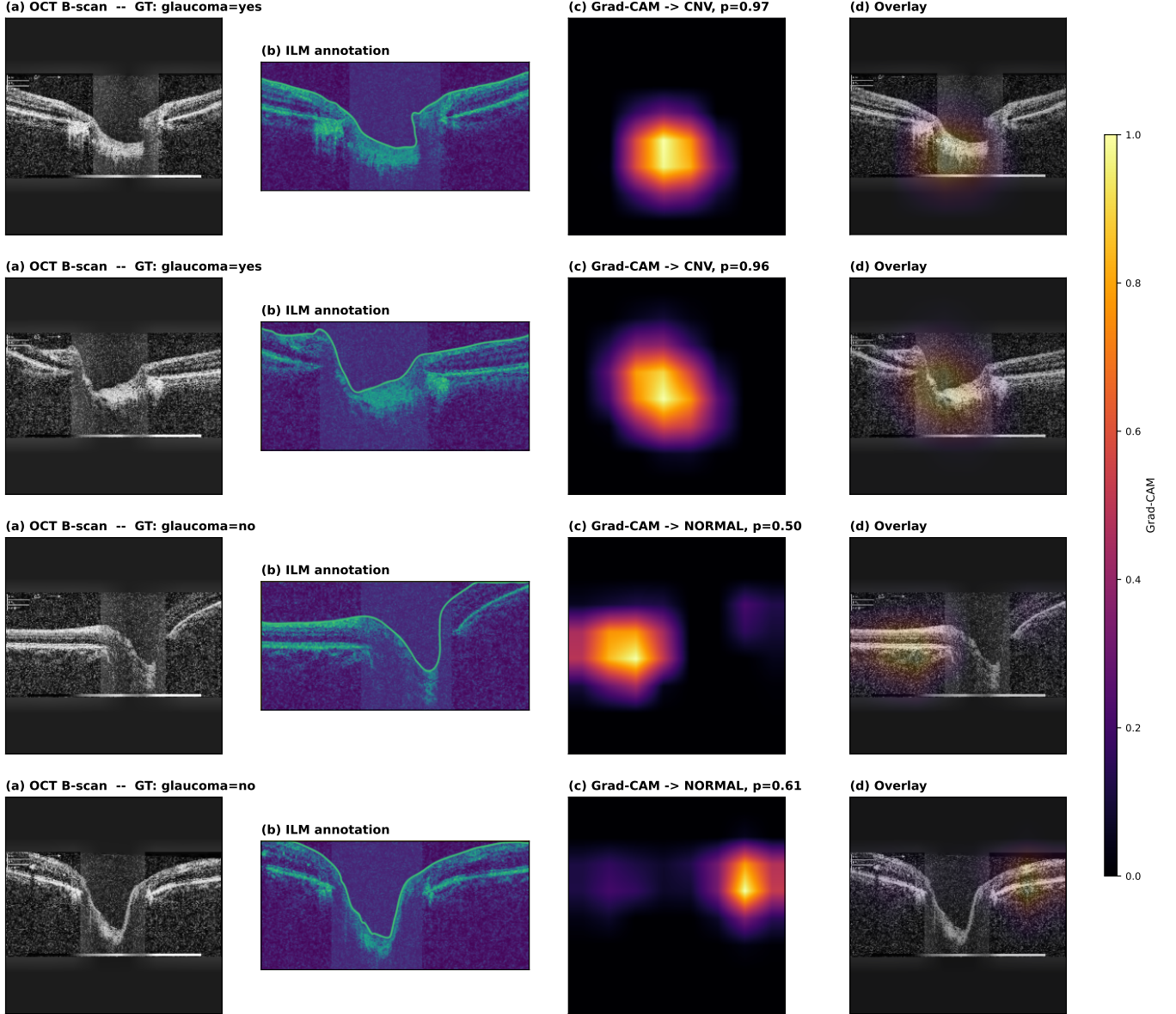


Figure 6: Qualitative paired-modality panel on the held-out ekacare glaucoma OCT cohort. (a) OCT B-scan with binary glaucoma label. (b) Cohort-provided ILM annotation. (c) Predicted-class Grad-CAM (Kermany class) on the same B-scan. (d) Overlay of (c) on (a). The activation overlays plausibly on annotated retinal layers despite the cross-cohort label-space mismatch.